

Advanced CPU Architecture & Hardware Accelerators

LAB2 preparation report

Synchronous HDL design - VHDL part2

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1. Aim of the Laboratory

- Obtaining skills in Synchronous HDL design (VHDL part 2) code, which contains Sequential and Behavioral code modeling.
- Obtaining skills in ModelSim (multi-language HDL simulation environment).
- Knowledge in digital systems design.
- Proper analysis and understanding of architectural design.

2. System Design definition

In this laboratory, you will design a synchronous digital system that detects valid sub-series for a given conditional value (see Table 1). The system block diagram is depicted in Figure 1. You are required to design the whole system and build a test bench.

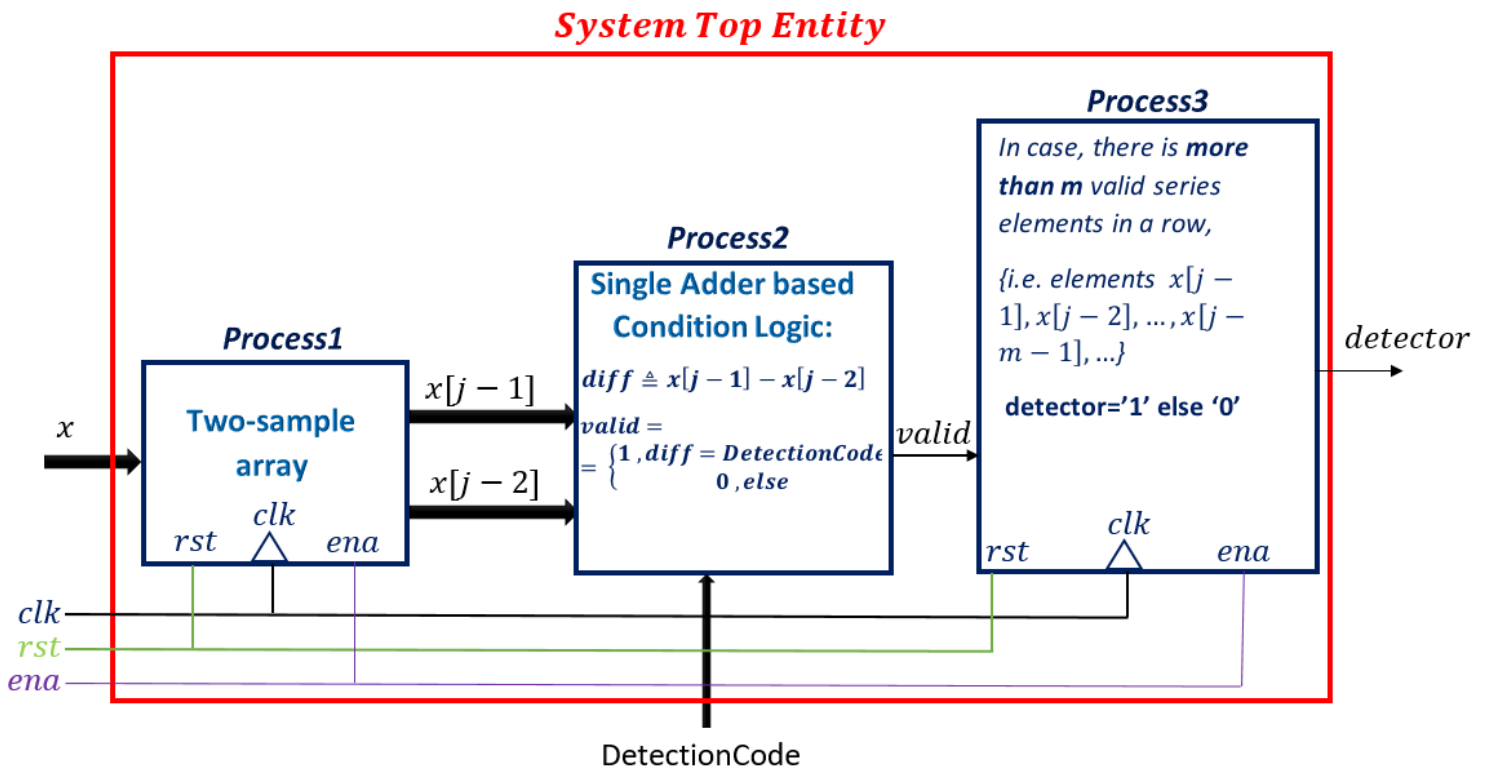
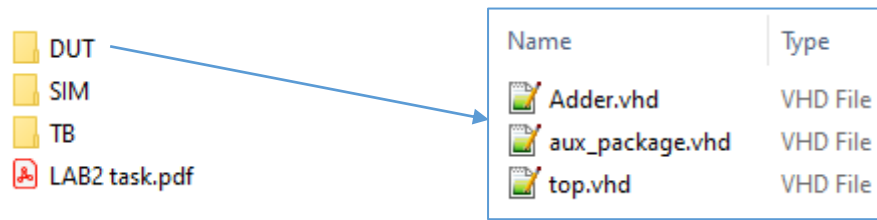


Figure 1 : System structure

<i>Detection Code</i>	<i>Condition</i>
0	$x[j-1] - x[j-2] = 1$
1	$x[j-1] - x[j-2] = 2$
2	$x[j-1] - x[j-2] = 3$
3	$x[j-1] - x[j-2] = 4$

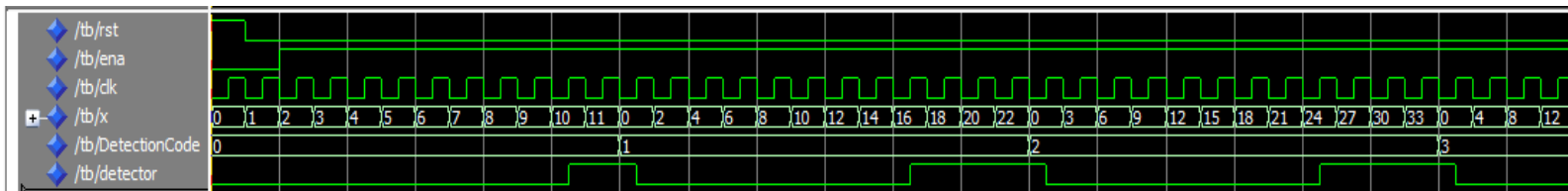
Table 1: cond (condition) values

- The Top-Level design must be Structural (*your DUT must contain the exact next three *.vhd files*).

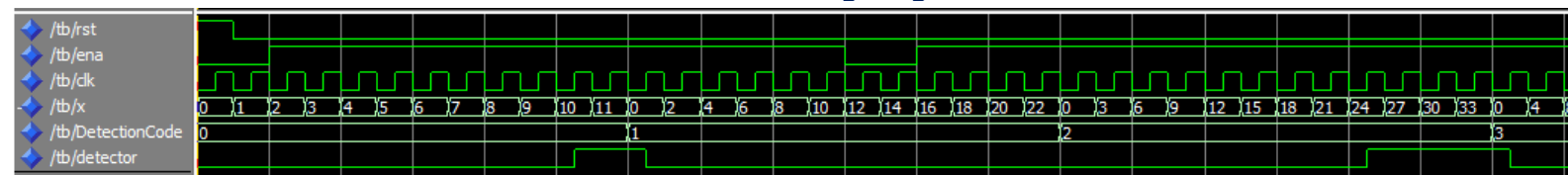


- The Top-Level design modeling can be Behavioral except the **Adder** based condition check module (single instantiation only of **Adder.vhd**). The main reason of this constraint is to minimize HW.
- **Note:** in the next three files that you must use in your project:
top.vhd, **aux_package.vhd**, **Adder.vhd** (you can only add your code to **top.vhd** file; you are not allowed to erase anything).
- The submitted project must be compiled using the given **tb1.vhd**, **tb2.vhd** files (otherwise the submitted project will be considered a failure).
- The submitted assignments get through copy checking machine, in this case, both sides' assignments will be disqualified.
- **Examples:**

tb_1 timing diagram



tb_2 timing diagram



tb_3 timing diagram

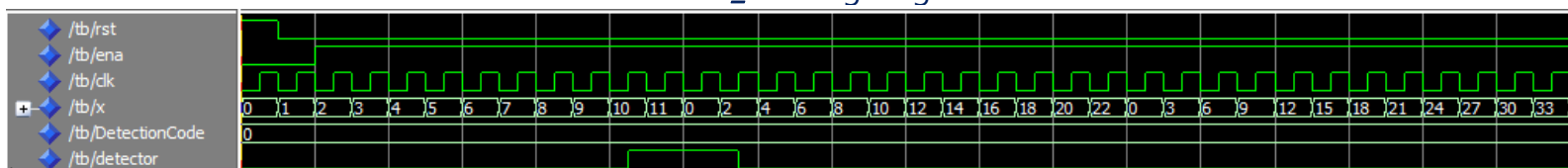
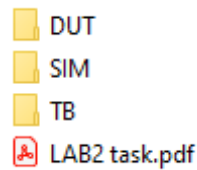
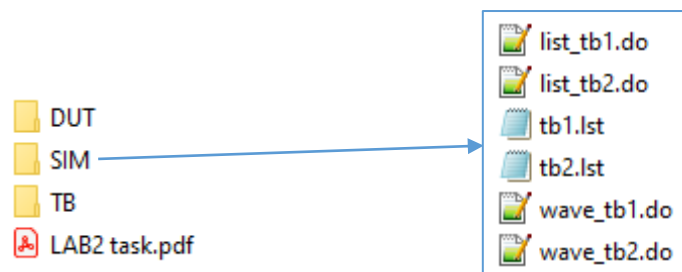


Figure 2: examples using waveforms

In addition, you are given two test bench files *tb1.vhd*, *tb2.vhd* (in TB folder) and their associate do and list files (in SIM folder).

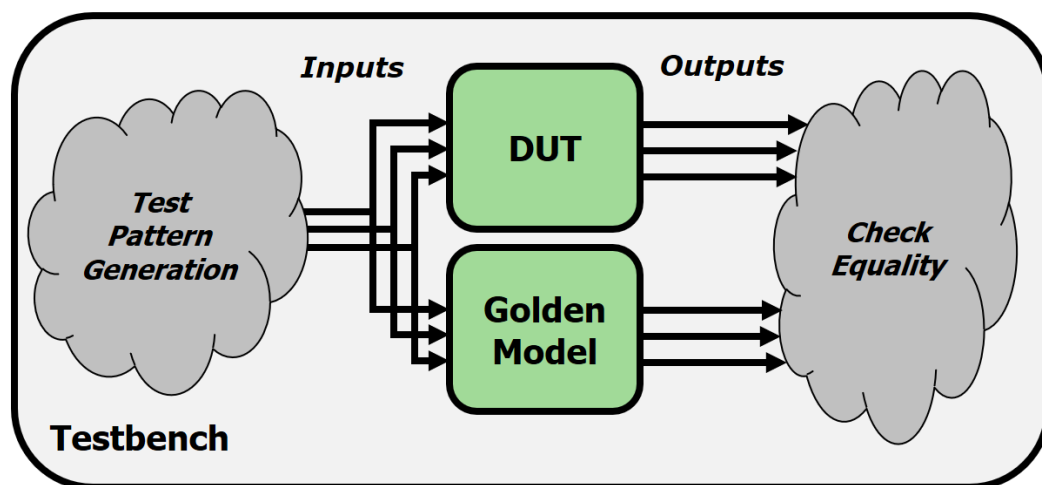


In order to use golden model based functional verification, you are given [TextDiff](#) application (download and double click the *TextDiff.exe* file) in order to compare your developing design results to the golden model results as part of design developing chain.



Automatic Testbench

The DUT **output** is compared against the **golden model**



Note: in comparison between the given *tb1.lst*, *tb2.lst* files and yours, you should ignore the *delta cycle* column

3. Test and Timing:

- Design a test bench, which tests all the system.
- Analyze the results by zooming on the important transactions in the waveforms.
Explain these (input/output/internal signals of the system).
- You are welcome to use the Internet also as reference.
- The timing of the system will be ideal (means a functional simulation).

4. Requirements

1. The design must be well commented on.
2. **Important:** For each of two submodules:
 - Graphical description (a square with ports going in and out) and short descriptions.
3. Elaborated analysis and wave forms:
 - Remove irrelevant signals.
 - Zoom on regions of interest.
 - Draw clouds on the waveform with explanations of what is happening (Figure 4).
 - Change the waveform colors in ModelSim for clear documentation
(Tools->Edit Preferences->Wave Windows).
4. A ZIP file in the form of **id1_id2.zip** (where id1 and id2 are the identification number of the submitters, and $id1 < id2$) *must be upload to Moodle only by student with id1* (any of these rules violation disqualify the task submission).

5. The **ZIP** file will contain (*only the exact next sub folders*):

Directory	Contains	Comments
DUT	Project VHDL files	Only VHDL files of DUT , excluding test bench Note: your project files must be well compiled without errors as a basic condition before submission
TB	VHDL files that are used for test bench	A single presentative <i>tb.vhd</i> (your version) of system top verification
SIM	DO files of <u>wave</u> and <u>list</u> forms	Do files for <i>tb.vhd</i> (your version) of system top verification
DOC	Project documentation	Readme.txt and pre2.pdf report file

Table 2: Directory Structure

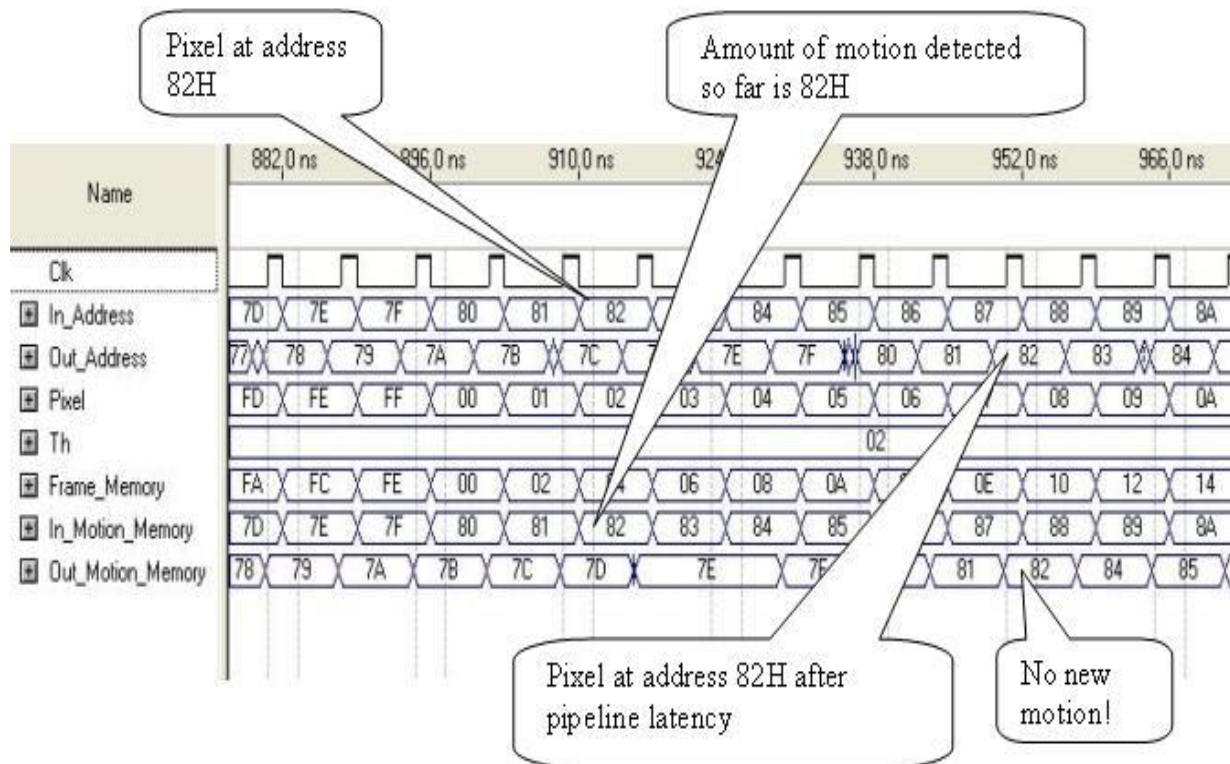
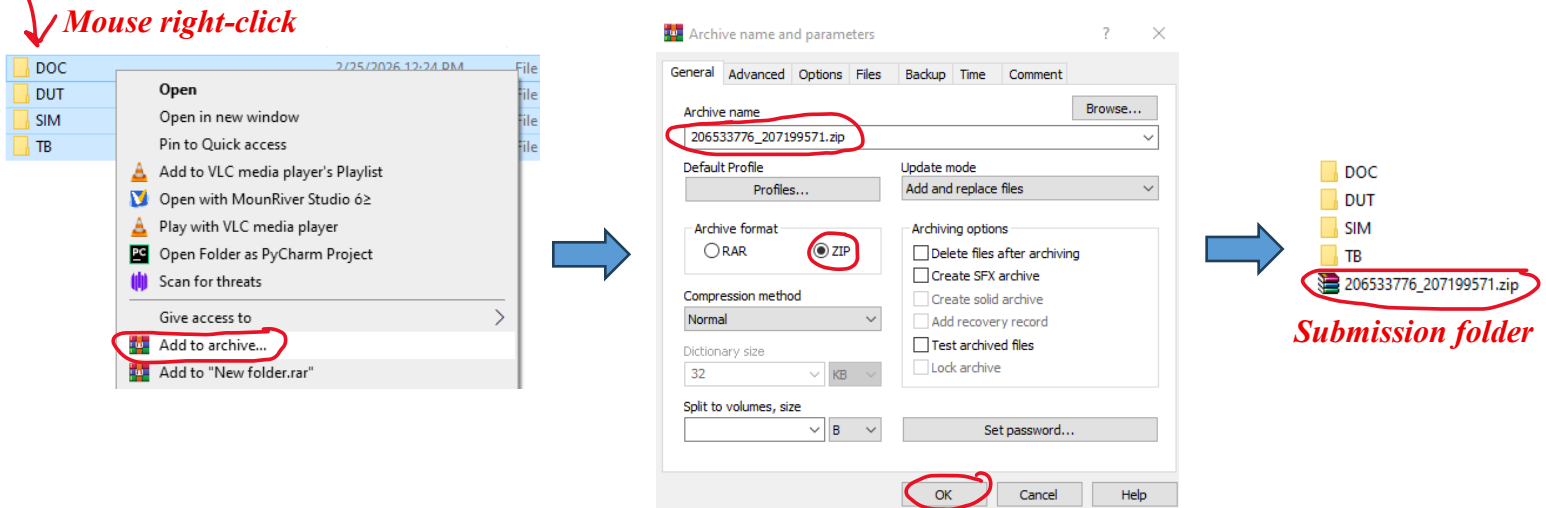


Figure 3: Clouds over the waveform example

5. Grading Policy

Weight	Task	Description
3%	Correct submission	The required four folders are the immediate subfolders under the submission id1 id2.zip folder
7%	Documentation	The “clear” way in which you presented the requirements, the analysis, and the conclusions on the work you’ve done
90%	Analysis and Test	The correct analysis of the system (under the requirements)

Table 1 : Grading



Under the above policies, you’ll also be evaluated using common sense:

- Your files will be compiled and checked; the system design must work.

Penalty Points	Penalty	Description
40	Uncompiled system design (1 st error type)	The submitted system design is non-compilable. Solution: An appeal submission is required, containing an elaboration of which DUT’s files and specific lines of code should be fixed for resubmission.
25	Simulation data static load error (2 nd error type)	The submitted system design is compilable, but has simulation static data load errors Solution: An appeal submission is required, containing an elaboration of which DUT’s files and specific lines of code should be fixed for resubmission.
20	Simulation runtime error (3 rd error type)	The submitted system design is compilable, but has simulation runtime errors Solution: An appeal submission is required, containing an elaboration of which DUT’s files and specific lines of code should be fixed for resubmission.

- Your design and architecture must be intelligent, minimal, effective, and well organized.